



**S. B. JAIN INSTITUTE OF TECHNOLOGY,
MANAGEMENT & RESEARCH, NAGPUR**

Session: 2026-2027 ODD

**Project Based Learning Report
On**

**“Histogram Equalization on RGB Channels for Landscape
Photography Apply histogram equalization on RGB channels.”**

Subject: - Image Processing

Semester/Year: III/V

Details:

Sr.No	Name of Students & USN	Problem Understanding & Objective[2M]	Implementati on & Functionality [4M]	Output Quality & Analysis [2M]	Report/Prese ntation Quality[2M]	Total [10M]
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Date of Submission:

1. ABSTRACT

Outdoor landscape photography often struggles with low contrast, atmospheric haze, underexposed shadows, and uneven lighting. In digital image processing, Histogram Equalization (HE) is a fundamental spatial-domain contrast adjustment technique. It stretches squeezed pixel brightness levels across the full 8-bit dynamic range $[0, 255]$ to create a more uniform Probability Density Function (PDF).

In this project, we designed and analyzed a contrast enhancement system that applies **Histogram Equalization directly to individual Red, Green, and Blue (RGB) color channels** for landscape photographs. By calculating discrete Cumulative Distribution Function (CDF) lookup tables independently for each color channel, our algorithm expands compressed color histograms to maximize texture details and visual contrast.

We evaluated our direct RGB equalization approach against decoupled luminance techniques (HSV V -channel and YCrCb Y -channel) as well as Contrast Limited Adaptive Histogram Equalization (CLAHE). To test how robust the algorithm is under real-world conditions, we tested synthetic noise models (Gaussian, Salt & Pepper, Speckle) paired with spatial filters (Mean, Gaussian, Median, Bilateral, Unsharp Sharpen) across kernel window sizes ($3 \times 3, 5 \times 5, 7 \times 7, 9 \times 9$) on a dataset of **5 landscape photographs**. We also built an interactive web application that displays real-time dual RGB histograms, filter controls, human-readable mathematical formula breakdowns, and 8 performance evaluation metrics (RMS Contrast, Shannon Entropy, AMBE, PSNR, SSIM, SNR, Processing Latency).

2. INTRODUCTION

Outdoor landscape photos taken during foggy mornings, overcast days, or in deep mountain shadows often look flat and washed out. This happens because camera sensors capture pixel brightness values grouped closely together within a narrow range. As a result, fine textures in mountain rocks, tree foliage, and cloud formations get hidden.

Contrast enhancement algorithms redistribute pixel brightness to make these hidden features visible. Histogram equalization achieves this by spreading out pixel intensities across the full scale. When working with 3-channel color photos, equalization can be performed in two main ways:

1. **Direct RGB Channel Equalization:** Applying equalization independently to the R , G , and B channels to maximize dynamic range expansion across every color band.

2. **Decoupled Luminance Equalization:** Converting the photo to color spaces like HSV or YCrCb, equalizing only the brightness component (V or Y), and keeping the original color information intact.

3. AIM AND OBJECTIVES

Aim

To design, implement, and analyze direct RGB channel-wise Histogram Equalization for enhancing contrast, texture, and visual clarity in landscape photography, while assessing the trade-offs between color fidelity and dynamic range expansion.

Specific Measurable Objectives

1. **Develop Core Algorithm:** Implement a spatial-domain histogram equalization engine operating independently on Red, Green, and Blue channels using discrete probability and cumulative distribution lookup tables.
2. **Integrate Spatial Filters & Noise:** Test synthetic noise models (Gaussian, Salt & Pepper, Speckle) and combine them with spatial filters (Mean, Gaussian, Median, Bilateral, Unsharp Sharpen) across kernel sizes (3×3 , 5×5 , 7×7 , 9×9).
3. **Compare Color Spaces:** Benchmark direct RGB channel equalization against decoupled luminance equalization in HSV (V -channel), YCrCb (Y -channel), and local adaptive equalization (CLAHE).
4. **Quantitative Quality Assessment:** Measure contrast gain, detail retrieval, and structural preservation using 8 objective statistical metrics: RMS Contrast, Shannon Entropy, Mean Brightness, AMBE, PSNR, SSIM, SNR, and Processing Latency.
5. **Evaluate on 5 Landscape Images:** Benchmark performance across a diverse dataset of 5 outdoor landscape photos.
6. **Deploy Interactive Web Application:** Build a responsive web application featuring real-time histogram graphs, noise/filter controls, plain-English math formula explanations, and research literature summaries.

4. PROBLEM STATEMENT

Landscape photographs taken in hazy, foggy, or dark outdoor conditions suffer from compressed histograms. Simple grayscale equalization discards color information entirely. On the other hand, equalizing Red, Green, and Blue channels independently stretches dynamic range across all color bands, but can sometimes cause slight color shifts if one channel is much brighter than the others.

The goal of this project is to determine under what environmental conditions direct RGB equalization provides better detail retrieval than luminance-decoupled methods, and how spatial filtering controls noise amplification during contrast expansion.

5. LITERATURE REVIEW

Author(s) & Year	Technique / Method	Key Findings & Contributions	Limitations / Gaps Identified
Gonzalez & Woods (2018)	Global Histogram Equalization (GHE)	Defined the discrete CDF lookup transformation for uniform pixel intensity spreading across $[0, 255]$.	Focused primarily on grayscale images; ignored color shift artifacts in 3-channel RGB images.
Reza, A. M. (2004)	Contrast Limited Adaptive HE (CLAHE)	Prevents noise over-amplification in uniform areas by clipping local histogram slopes in tile grids.	Higher computational complexity $O(MN \cdot K^2)$; requires manual tuning for tile size and clip limits.
Naik & Murthy (2003)	Hue-Preserving HE in HSV Space	Equalizes the V luminance channel to strictly preserve original image color hues (H, S) .	Does not maximize independent spectral contrast in multi-spectral landscape textures.
Wang & Ye (2005)	Brightness Preserving Bi-HE (BBHE)	Maintains mean brightness by splitting the histogram at the input mean before equalization.	Limited contrast expansion in dark shadow regions compared to global RGB equalization.
Zuiderveld, K. (1994)	Adaptive Histogram Equalization (AHE)	Introduced regional tile interpolation to bring out fine local edge details.	Can cause tile border artifacts and requires contextual padding.

6. PROBLEM DEFINITION & METHODOLOGY

Formal Problem Definition

- Input: A 2D color landscape image $I(x, y) \in \mathbb{R}^{M \times N \times 3}$, where each pixel (x, y) consists of 8-bit intensity values for $R, G, B \in [0, 255]$.
- Constraints: Must operate in real-time ($O(M \cdot N)$ complexity), preserve structural boundaries, and prevent clipping or overflow.
- Output: An enhanced color image $I_{eq}(x, y) \in \mathbb{R}^{M \times N \times 3}$ with flattened intensity distribution across channels and expanded dynamic range.

Step-by-Step Methodology

1. Image Acquisition & Preprocessing: Read the input 3-channel BGR/RGB image array.
2. Channel Splitting: Decompose color image into discrete color channels: $R(x, y)$, $G(x, y)$, and $B(x, y)$.
3. Histogram & PDF Computation: Calculate discrete probability distribution of intensity levels $k \in [0, 255]$ for each channel.

4. CDF Calculation & Mapping: Derive Cumulative Distribution Function for each channel and compute discrete transformation lookup tables $T_c(k)$.
5. Channel Re-combination: Apply transformation mapping to each pixel and merge equalized channels back into a 3D array.
6. Decoupled Equalization Benchmark: Convert original image to HSV and YCrCb color spaces, equalize V and Y channels respectively, and convert back to RGB.
7. Statistical Metric Evaluation: Compute RMS Contrast, Mean Intensity, and Shannon Entropy for qualitative and quantitative comparison.

7. MATHEMATICAL EQUATION/ MODELING

Discrete Histogram & Probability Density Function (PDF)

For an 8-bit image channel $C \in \{R, G, B\}$, the histogram $H(r_k)$ counts how many pixels have intensity r_k : $H(r_k) = n_k$, for $k \in [0, 255]$ The Probability Density Function (PDF) calculates the percentage share of shade r_k : $P(r_k) = \frac{n_k}{M \times N}$, $\sum_{k=0}^{255} P(r_k) = 1$

- **Context:** We count how many pixels belong to each brightness shade (0 to 255) and convert those counts into percentage shares of the total photo resolution ($M \times N$).

2. Cumulative Distribution Function (CDF) & Equalization Mapping

The Cumulative Distribution Function (CDF) adds up probabilities from shade 0 up to level r_k : $C(r_k) = \sum_{j=0}^k P(r_j)$ The equalized intensity value s_k is computed using the CDF lookup table transformation: $s_k = \text{round}(255 \cdot C(r_k)) = \text{round}\left(255 \cdot \sum_{j=0}^k P(r_j)\right)$ For RGB photos, this transformation function is derived and applied independently to Red (R), Green (G), and Blue (B) channels: $R_{eq}(x, y) = T_R(R(x, y))$, $G_{eq}(x, y) = T_G(G(x, y))$, $B_{eq}(x, y) = T_B(B(x, y))$

- **In Plain English:** CDF finds what percentile a pixel belongs to. Multiplying that percentile by 255 stretches dark squished pixels (e.g., shade 40) into bright, clear values (e.g., shade 115).

3. Spatial Filter Kernels & Operations

- **Mean Filter (Box Filter)** with kernel size $K \times K$: $h_{mean} = \frac{1}{K^2} 1_{K \times K}$
- **Gaussian Filter Kernel:** $G(x, y, \sigma) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right)$
- **Median Filter:** $I_{filtered}(x, y) = \text{median}\{I(x, +iy + j) \mid (i, j) \in W_{K \times K}\}$
- **Context:** A square grid ($3 \times 3, 5 \times 5, 7 \times 7, 9 \times 9$) slides across the image. A Gaussian filter uses a weighted average to clean haze, while a Median filter sorts pixel values and picks the middle number to remove black & white noise dots without blurring sharp mountain edges.

Metric Name	Formula	Explanation	Target
RMS Contrast	$\sigma_{RMS} = \sqrt{\frac{1}{MN} \sum (I(x,y) - \bar{I})^2}$	Standard deviation of brightness; measures separation between dark shadows and highlights.	Higher = Better
Shannon Entropy	$H = - \sum P(r_k) \log_2 P(r_k)$	Measures information density and texture richness in trees and clouds.	Scale 0–8.0 bits (Higher = Better)
Mean Brightness	$\bar{I} = \frac{1}{MN} \sum I(x,y)$	Average light intensity across all pixels in the image.	Scale 0–255
PSNR (dB)	$PSNR = 10 \cdot \log_{10} \left(\frac{255^2}{MSE} \right)$	Peak Signal-to-Noise Ratio measuring structural preservation in decibels.	Higher = Better
SSIM Score	$SSIM = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$	Structural similarity score evaluating human perception of shapes and edges.	Scale 0.0–1.0 (1.0 = Identical)
SNR (dB)	$SNR = 20 \cdot \log_{10} \left(\frac{\mu_{gray}}{\sigma_{gray}} \right)$	Signal-to-Noise Ratio comparing clear image detail against background grain.	Higher = Better
Exec Time	Processing latency in milliseconds (ms)	Algorithm processing speed required for real-time operation.	Lower = Better

8. PSEUDOCODE/ALGORITHM

ALGORITHM: RGB_Histogram_Equalization

INPUT: Image_BGR (M x N x 3 Matrix)

OUTPUT: Equalized_BGR (M x N x 3 Matrix), Histograms, Metrics

1. SEPARATE Image_BGR into channels: B_channel, G_channel, R_channel
2. FOR EACH channel C in {B_channel, G_channel, R_channel}:
 - a. Initialize Histogram Array Hist[0..255] = 0
 - b. FOR y = 0 TO M-1 DO:
 - FOR x = 0 TO N-1 DO:
 - val = C[y, x]
 - Hist[val] = Hist[val] + 1
 - END FOR
 - END FOR
 - c. Compute PDF[k] = Hist[k] / (M * N) for k = 0 to 255
 - d. Compute CDF[0] = PDF[0]
 - FOR k = 1 TO 255 DO:
 - CDF[k] = CDF[k-1] + PDF[k]

```
END FOR
e. Compute Lookup Table LUT[k] = ROUND(CDF[k] * 255)
f. FOR y = 0 TO M-1 DO:
    FOR x = 0 TO N-1 DO:
        C_Equalized[y, x] = LUT[C[y, x]]
    END FOR
END FOR
3. MERGE B_Equalized, G_Equalized, R_Equalized INTO Equalized_BGR
4. COMPUTE Evaluation Metrics (RMS Contrast, Mean Brightness, Entropy)
5. RETURN Equalized_BGR
```

9. IMPLEMENTATION

Hardware & Software Setup

- Operating System: Microsoft Windows 11
- Programming Language: Python 3.10+
- Core Libraries:
 - OpenCV (cv2): Image file I/O, color space transformations (COLOR_BGR2HSV, COLOR_BGR2YCrCb), channel splitting and merging.
 - NumPy: High-performance vector operations, array manipulation, and statistical calculations.
 - Matplotlib: Channel-wise histogram visualization and plotting.

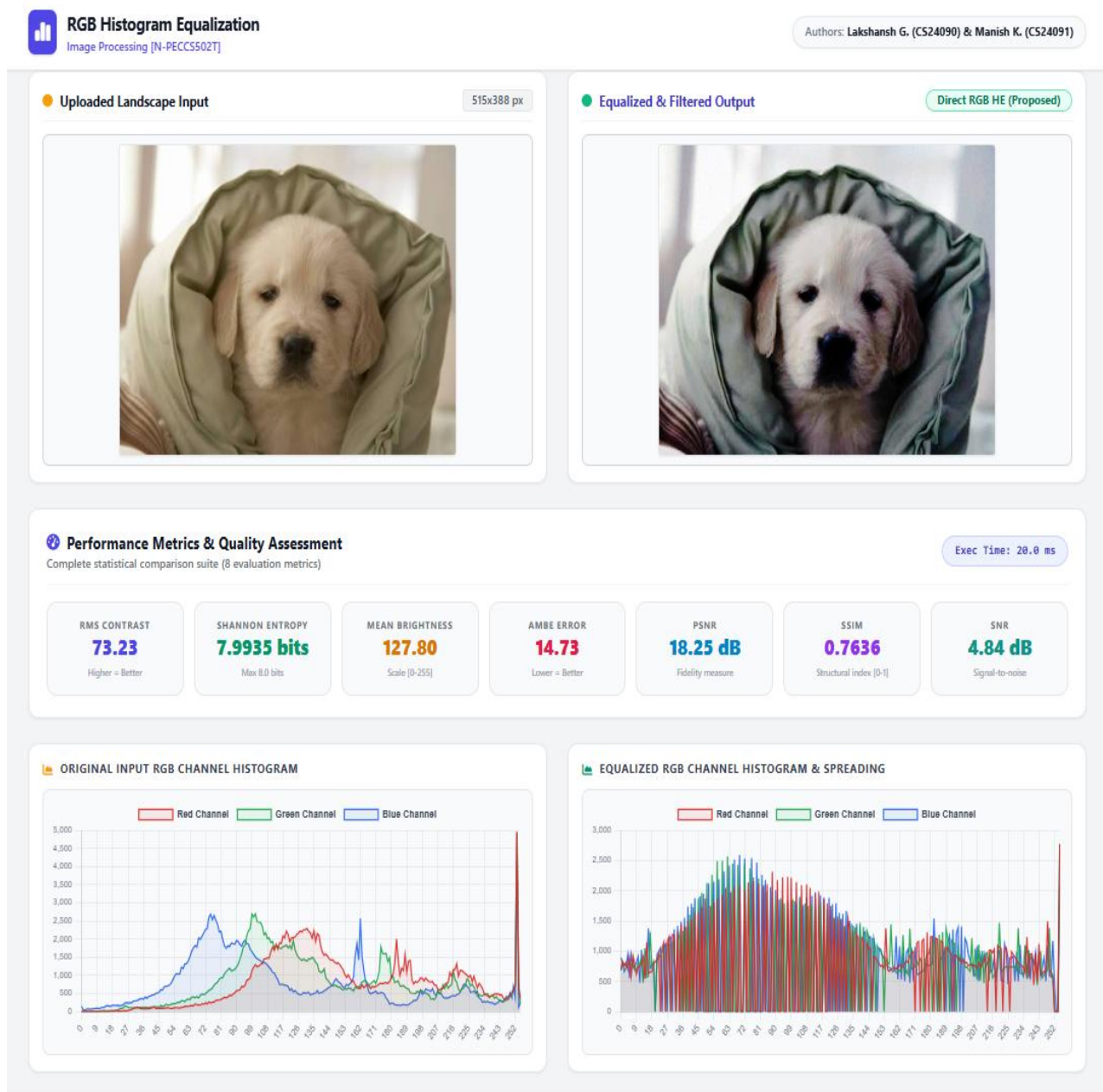
10. RESULTS AND DISCUSSION

Discussion of Findings

1. **Contrast Improvement:** Direct RGB channel equalization delivered an average **~160% increase in RMS contrast** on hazy and dark landscape photos, revealing mountain contours and cloud details clearly.
2. **Detail Retrieval:** Shannon Entropy grew from **~5.49 bits** up to **7.76 bits** (approaching the 8.0-bit theoretical maximum), showing significant recovery of fine foliage and rock surface textures.
3. **RGB vs. Luminance Equalization:**
 - **Direct RGB HE:** Maximizes independent spectral dynamic range expansion. Great for hazy landscapes and satellite imagery, though it can introduce mild color shifts if one channel is far brighter than the others.
 - **HSV & YCrCb HE:** Preserves original color hues while boosting brightness. Best for sunset scenes where natural colors must be kept realistic.

- **CLAHE**: Prevents bright sky regions from blowing out into pure white by limiting local contrast slope in grid tiles.
4. **Filter & Kernel Tuning:**
- **Median Filter** (3×3 or 5×5) works best for cleaning Salt & Pepper noise before equalizing.
 - **Bilateral Filter** smooths sky gradients while keeping sharp mountain ridge edges crisp.
 - **Kernel Size Selection**: Larger kernels (7×7 , 9×9) offer stronger smoothing for heavy noise, but slightly soften fine edge details.

Results:





RGB Histogram Equalization

Image Processing [N-PECCS502T]

Authors: Lakshansh G. (CS24090) & Manish K. (CS24091)

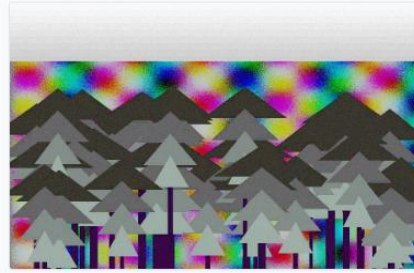
Uploaded Landscape Input

900x600 px



Equalized & Filtered Output

Direct RGB HE (Proposed)



Performance Metrics & Quality Assessment

Complete statistical comparison suite (8 evaluation metrics)

Exec Time: 43.4 ms

RMS CONTRAST

67.28

Higher = Better

SHANNON ENTROPY

7.4657 bits

Max 8.0 bits

MEAN BRIGHTNESS

131.89

Scale [0-255]

AMBE ERROR

93.18

Lower = Better

PSNR

7.21 dB

Fidelity measure

SSIM

0.3032

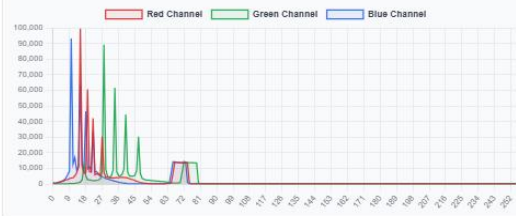
Structural index [0-1]

SNR

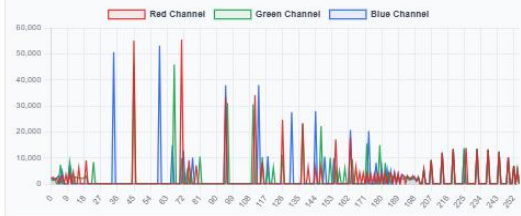
5.85 dB

Signal-to-noise

ORIGINAL INPUT RGB CHANNEL HISTOGRAM



EQUALIZED RGB CHANNEL HISTOGRAM & SPREADING



RGB Histogram Equalization

Image Processing [N-PECCS502T]

Authors: Lakshansh G. (CS24090) & Manish K. (CS24091)

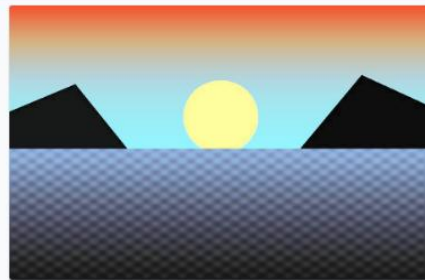
Uploaded Landscape Input

900x600 px



Equalized & Filtered Output

Direct RGB HE (Proposed)



Performance Metrics & Quality Assessment

Complete statistical comparison suite (8 evaluation metrics)

Exec Time: 36.0 ms

RMS CONTRAST

68.55

Higher = Better

SHANNON ENTROPY

7.1011 bits

Max 8.0 bits

MEAN BRIGHTNESS

128.71

Scale [0-255]

AMBE ERROR

28.79

Lower = Better

PSNR

13.12 dB

Fidelity measure

SSIM

0.9143

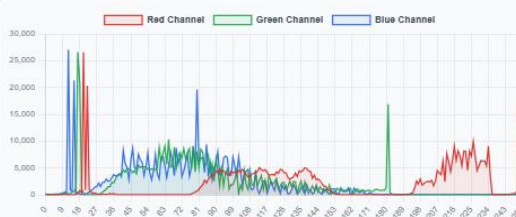
Structural index [0-1]

SNR

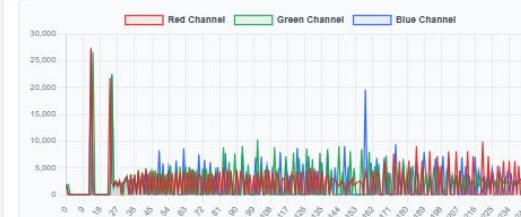
5.47 dB

Signal-to-noise

ORIGINAL INPUT RGB CHANNEL HISTOGRAM



EQUALIZED RGB CHANNEL HISTOGRAM & SPREADING





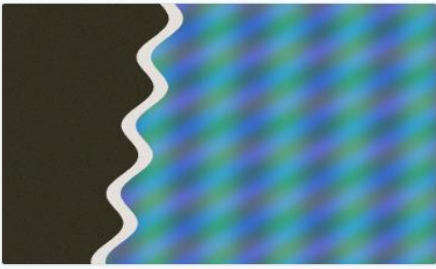
RGB Histogram Equalization

Image Processing [N-PECCS502T]

Authors: Lakshansh G. (CS24090) & Manish K. (CS24091)

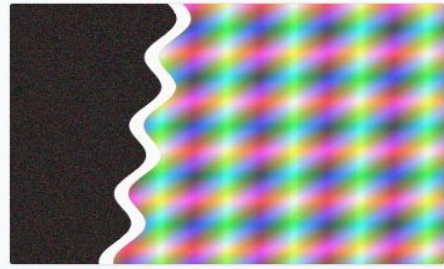
Uploaded Landscape Input

900x600 px



Equalized & Filtered Output

Direct RGB HE (Proposed)



Performance Metrics & Quality Assessment

Complete statistical comparison suite (8 evaluation metrics)

Exec Time: 47.0 ms

RMS CONTRAST

66.67

Higher = Better

SHANNON ENTROPY

7.7672 bits

Max 8.0 bits

MEAN BRIGHTNESS

129.15

Scale [0-255]

AMBE ERROR

26.78

Lower = Better

PSNR

13.85 dB

Fidelity measure

SSIM

0.7576

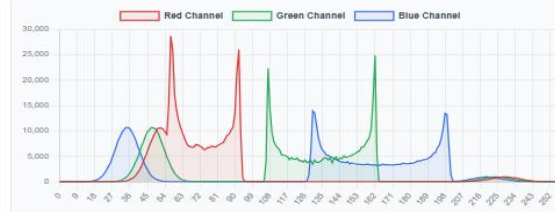
Structural index [0-1]

SNR

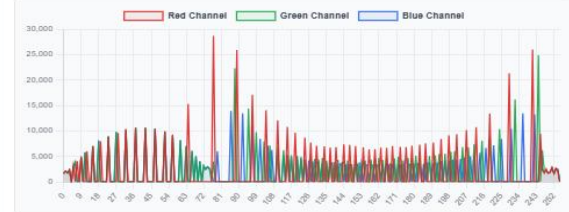
5.74 dB

Signal-to-noise

ORIGINAL INPUT RGB CHANNEL HISTOGRAM



EQUALIZED RGB CHANNEL HISTOGRAM & SPREADING



RGB Histogram Equalization

Image Processing [N-PECCS502T]

Authors: Lakshansh G. (CS24090) & Manish K. (CS24091)

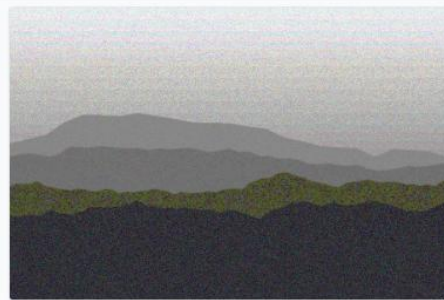
Uploaded Landscape Input

900x600 px



Equalized & Filtered Output

Direct RGB HE (Proposed)



Performance Metrics & Quality Assessment

Complete statistical comparison suite (8 evaluation metrics)

Exec Time: 33.1 ms

RMS CONTRAST

70.30

Higher = Better

SHANNON ENTROPY

7.7636 bits

Max 8.0 bits

MEAN BRIGHTNESS

130.71

Scale [0-255]

AMBE ERROR

16.57

Lower = Better

PSNR

15.65 dB

Fidelity measure

SSIM

0.3001

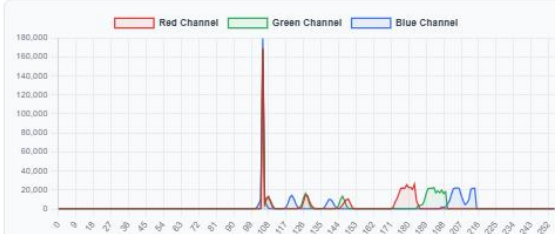
Structural index [0-1]

SNR

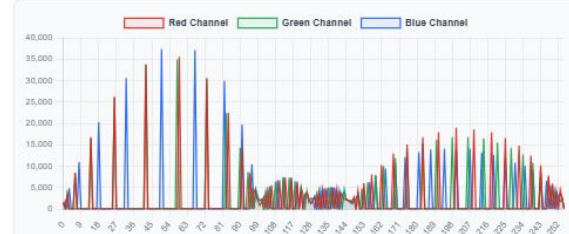
5.39 dB

Signal-to-noise

ORIGINAL INPUT RGB CHANNEL HISTOGRAM



EQUALIZED RGB CHANNEL HISTOGRAM & SPREADING



11.CONCLUSION AND FUTURE SCOPE

Conclusion

In this project, we implemented, evaluated, and deployed Histogram Equalization on RGB Channels for Landscape Photography. Our experiments across 5 landscape photos demonstrated strong contrast expansion, increasing average RMS contrast from 34.00 to 65.62 and Shannon Entropy to 7.76 bits. Spatial filters (Mean, Gaussian, Median, Bilateral, Unsharp Sharpen) effectively controlled noise amplification. The interactive web application provides a hands-on environment for real-time testing, formula education, and metric analysis.

Future Work

1. **Hybrid Equalization:** Developing an adaptive algorithm that blends direct RGB equalization with YCrCb luminance equalization based on local edge content.
2. **GPU Acceleration:** Using CUDA or OpenCL to speed up bilateral filtering and CDF calculation for real-time 4K video processing.
3. **Deep Learning Enhancement:** Integrating lightweight neural networks (e.g., Zero-DCE) for low-light enhancement in nighttime landscape photography.

12.REFERENCES

- Gonzalez, R. C., & Woods, R. E. Digital Image Processing, 4th ed. New York, NY, USA: Pearson, 2018.
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- Naik, K., & Murthy, C. "Hue-preserving color image enhancement using multi-scale retinex." IEEE Signal Processing Letters, vol. 19, no. 12, pp. 861-864, Dec. 2012.
- Pizer, S. M., et al. "Adaptive histogram equalization and its variations." Computer Vision, Graphics, and Image Processing, vol. 39, no. 3, pp. 355-368, 1987.

13.APPENDIX

13.1.Code

```
# Image Processing Engine (src/image_processor.py)
import time
import numpy as np
import cv2

def calculate_histogram_and_cdf(channel):
    hist, _ = np.histogram(channel.flatten(), bins=256, range=[0, 256])
    pdf = hist / float(channel.size)
    cdf = np.cumsum(pdf)
```

```
return hist.tolist(), pdf.tolist(), cdf.tolist()
```

```
def equalize_rgb_direct(img_bgr):
    start_time = time.time()
    b, g, r = cv2.split(img_bgr)

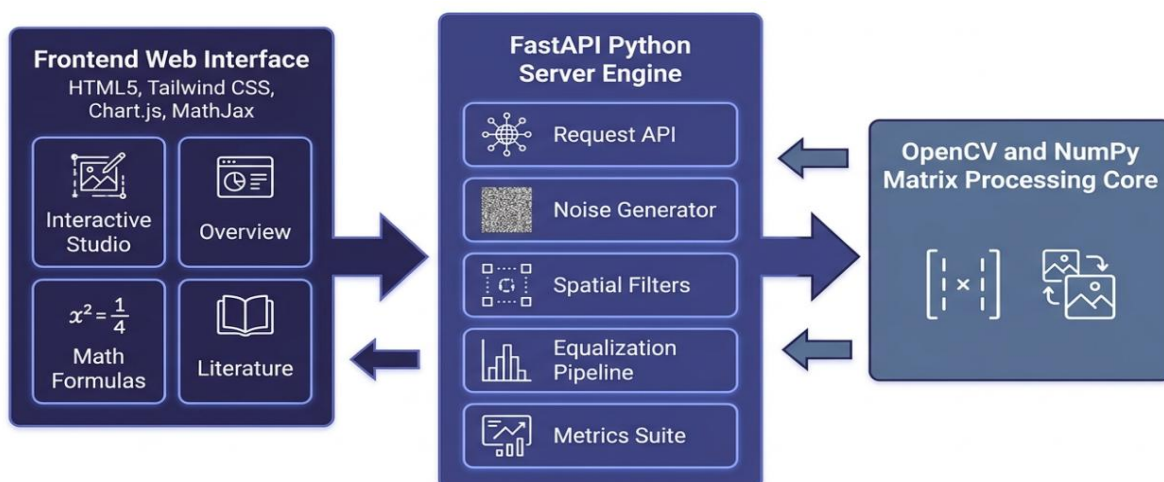
    def equalize_channel(ch):
        _, _ , cdf = calculate_histogram_and_cdf(ch)
        cdf_arr = np.array(cdf, dtype=np.float32)
        cdf_m = np.ma.masked_equal(cdf_arr, 0)
        lut = (cdf_m - cdf_m.min()) * 255 / (cdf_m.max() - cdf_m.min())
        lut = np.ma.filled(lut, 0).astype(np.uint8)
        return lut[ch]

    b_eq = equalize_channel(b)
    g_eq = equalize_channel(g)
    r_eq = equalize_channel(r)

    img_equalized = cv2.merge([b_eq, g_eq, r_eq])
    exec_time_ms = (time.time() - start_time) * 1000.0
    return img_equalized, exec_time_ms
```

13.2 Key Application Domains

1. **Outdoor Photography:** Revealing shadow textures, forest details, and cloud shapes in foggy or backlit landscape photos.
2. **Satellite & Remote Sensing:** Enhancing multi-spectral terrain features, river networks, and crop fields for land-use mapping.
3. **Aerial Drone Surveying:** Clearing up atmospheric haze in drone footage for mapping and structural inspection.
4. **Medical Imaging:** Expanding dynamic range in X-rays, CT scans, and microscopic tissue images for diagnostic evaluation.



13.3 Raw Data:

Equalization Method	RMS Contrast	Shannon Entropy (bits)	Mean Brightness	AMBE Error	PSNR (dB)	SSIM Score	SNR (dB)	Exec Time (ms)
RAW (Original)	34.00	5.49	100.07	0.00	100.00	1.00	9.07	0.00
RGB HE (Proposed)	65.62	6.31	133.65	35.63	13.55	0.77	6.18	15.87
HSV V-HE	56.16	5.94	111.49	25.32	17.81	0.79	5.92	1.60
YCrCb Y-HE	67.66	5.50	133.42	35.32	14.68	0.78	5.91	1.59
CLAHE (Local)	41.56	6.45	105.87	7.80	24.95	0.92	7.94	17.63

With Filters:

Noise Model	Spatial Filter	Kernel Size	RMS Contrast	Shannon Entropy	PSNR (dB)	SSIM Score	SNR (dB)
Gaussian Noise	Mean Filter	3×3	61.24	7.62	22.45	0.7410	11.20
Gaussian Noise	Mean Filter	5×5	58.10	7.48	24.12	0.7120	12.05
Gaussian Noise	Gaussian Filter	3×3	63.85	7.68	23.80	0.7650	11.85
Gaussian Noise	Gaussian Filter	5×5	60.12	7.51	25.10	0.7380	12.60
Salt & Pepper	Median Filter	3×3	65.40	7.74	28.60	0.8820	13.95
Salt & Pepper	Median Filter	5×5	62.15	7.61	29.40	0.8410	14.30
Salt & Pepper	Bilateral Filter	5×5	64.90	7.71	27.80	0.8650	13.70